Electronic Engineering - Embedded Systems

Lab 2 Report

Numerical analysis of MTLs

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1 Process description

1.1 Problem nr.1

The first part of the problem was to synthetize the microstrip, meaning that I proceeded with the calculations necessary in order to get all the parameters of the microstrip. Since the initial parameters are:

$$Z_\infty = 50\ \Omega \tag{1}$$

$$\epsilon_r = 9.8 \tag{2}$$

I checked what equation to use in order to get the value of $\frac{W}{h}$ and the value resulted to be inferior to 2: This means that I can use only the first equation to get the value that

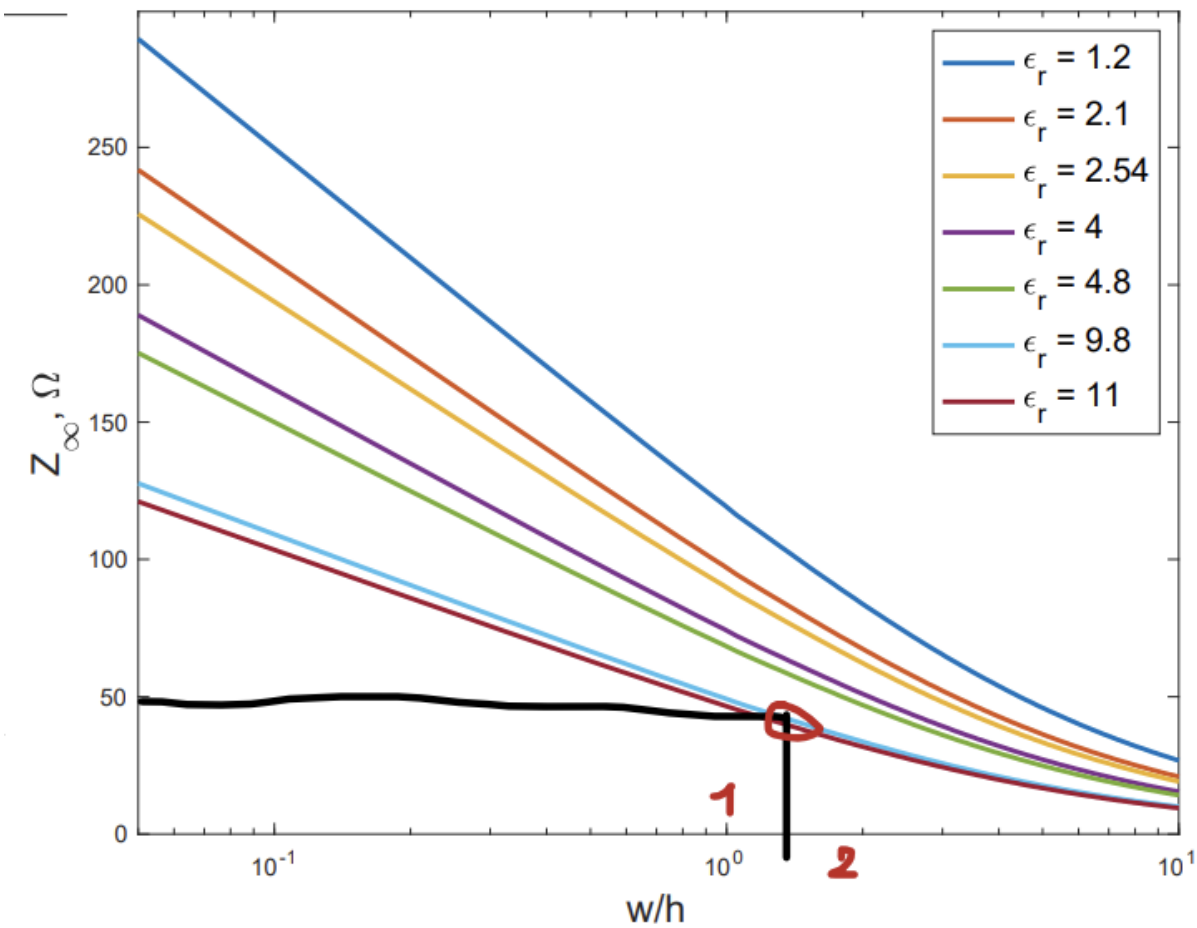


Figure 1: Variation of Z_∞ with $\frac{W}{h}$

I need, so I only had to compute the intermediate value A:

$$A = \frac{Z_\infty}{60}\sqrt{\frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{\varepsilon_r + 1}\left(0.23 + \frac{0.11}{\varepsilon_r}\right)} \tag{3}$$

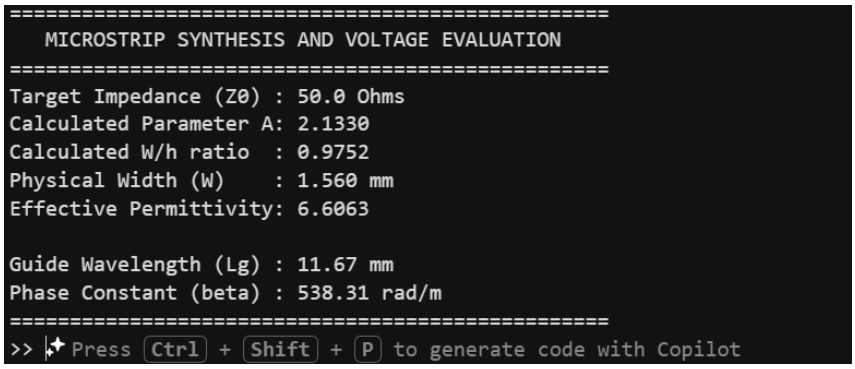
and with this $\frac{W}{h}$ is:

$$\frac{W}{h} = \frac{8e^A}{e^{2A} - 2} \tag{4}$$

Finally I used this value to calculate the effective dielectric constant, by using the following formula:

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \cdot \frac{1}{\sqrt{1 + 12h/W}} \tag{5}$$

For the last part of Problem nr.1 I computed the values of the guide wavelenght and the phase constant; this was needed because to evaluate the voltage along the line I have to use the telegrapher's equation. The results are:



```
====================================================
   MICROSTRIP SYNTHESIS AND VOLTAGE EVALUATION
====================================================
Target Impedance (Z0) : 50.0 Ohms
Calculated Parameter A: 2.1330
Calculated W/h ratio  : 0.9752
Physical Width (W)    : 1.560 mm
Effective Permittivity: 6.6063

Guide Wavelength (Lg) : 11.67 mm
Phase Constant (beta) : 538.31 rad/m
====================================================
>> Press Ctrl + Shift + P to generate code with Copilot
```

Figure 2: Results from MATLAB script

In the end I plotted everything to see how $V_{iso}(z)$ changes:

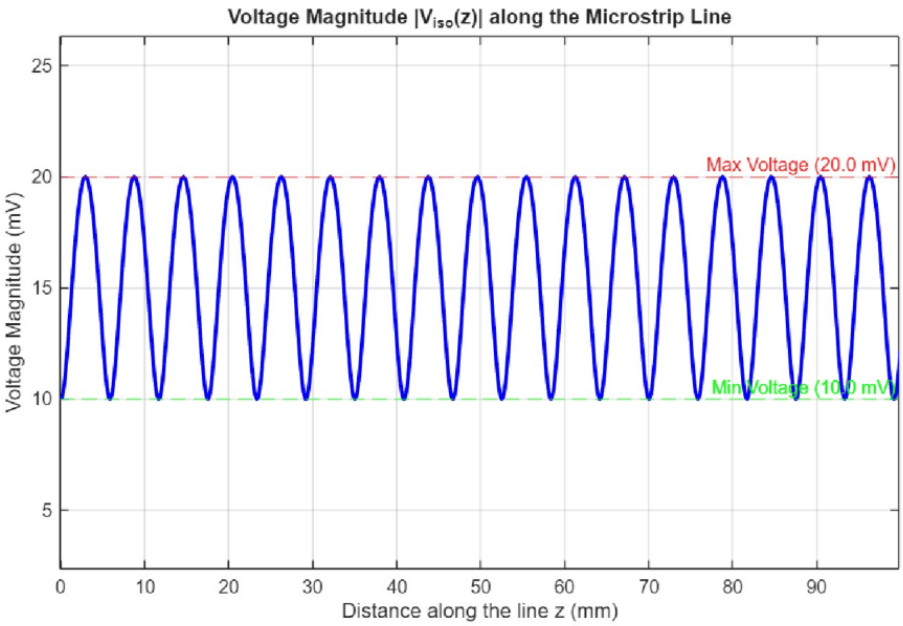


Figure 3: Voltage Magnitude $|V_{iso}(z)|$ along the Microstrip Line

1.2 Problem nr.2

By using the program I got the following results:

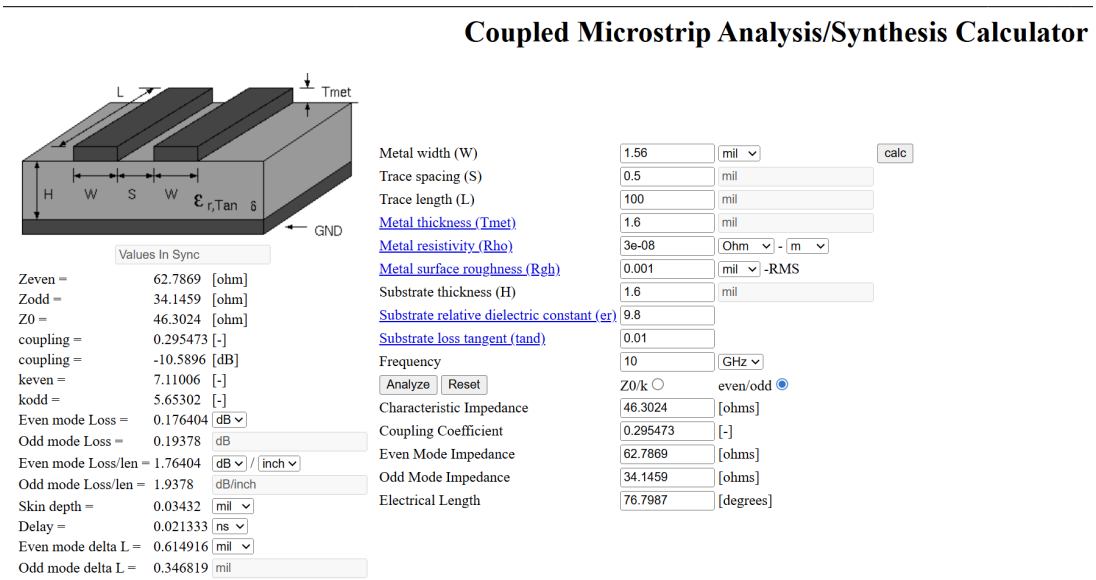


Figure 4: 0.5 mm spacing MTLs

The most important being: $Z_{even} = 62.7869\ \Omega$, $Z_{odd} = 34.1459\ \Omega$, $k_{even} = 7.11006$, and $k_{odd} = 5.65302$

The following step was to evaluate the true phase constants: β_{even} and β_{odd} and use them to calculate even and odd voltages and finally I used them to evaluate the final voltage. This was the result I got:

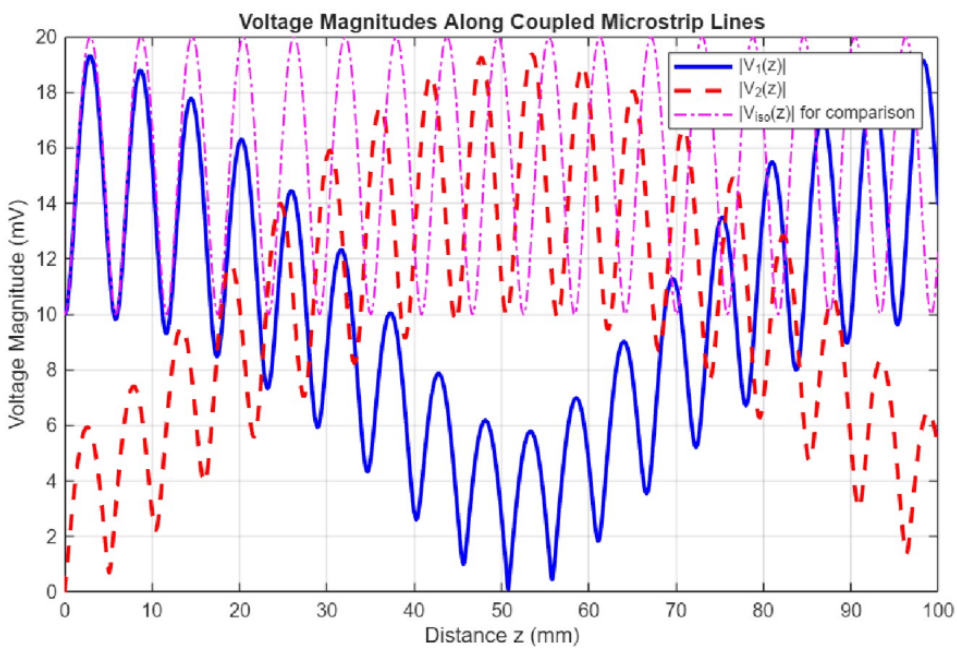


Figure 5: Voltage Magnitudes Along Coupled Microstrip Lines, s = 0.5 mm

From the data given in the problem, line 2 should be a ”silent line”, in fact no voltage nor current flow there; but due to the vicinity to another active line We can see that some energy from line 1 ”bleeds” into line 2 causing crosstalk phenomenon. Now we ran the exact same tests, but we increase the spacing to 10 mm.

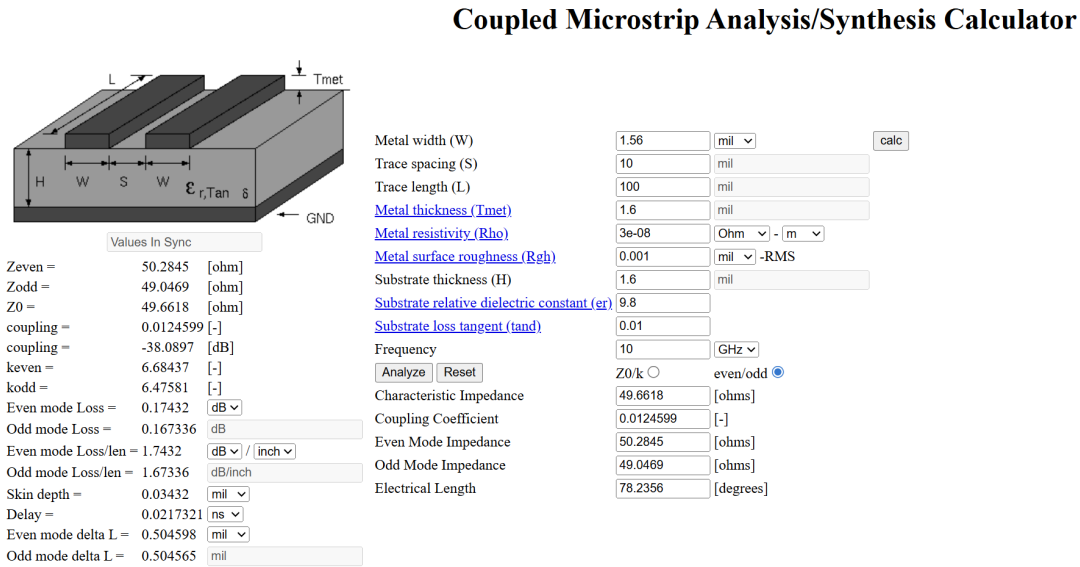


Figure 6: 10 mm spacing MTLs

and with this data we get:

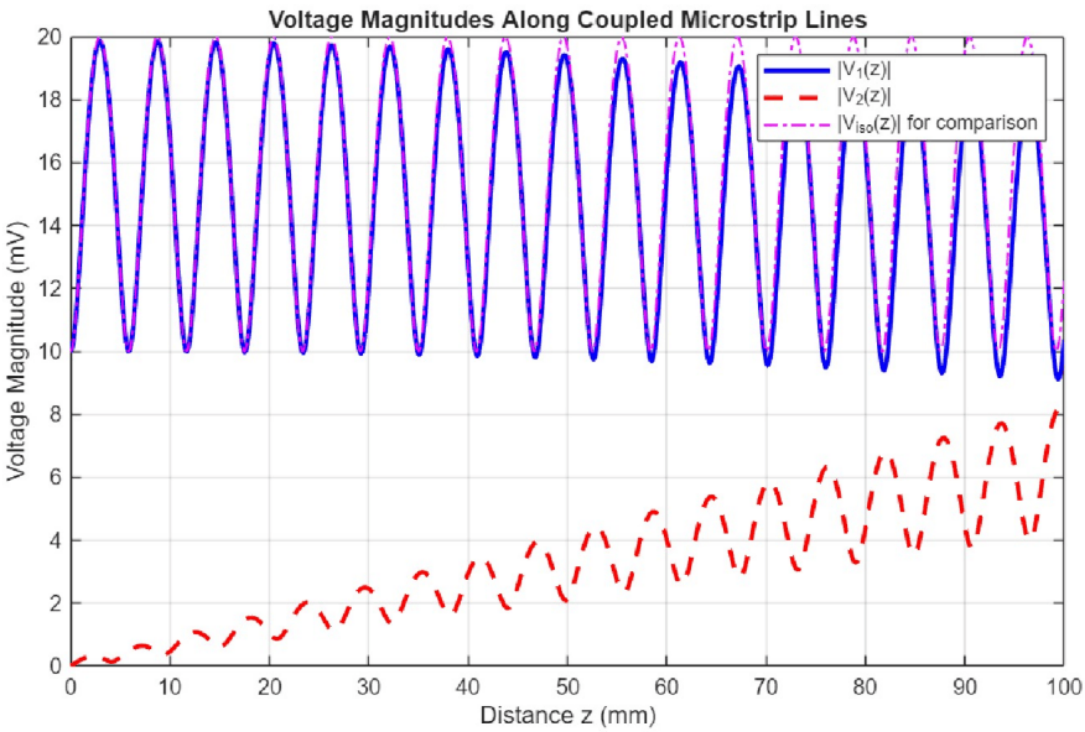


Figure 7: Voltage Magnitudes Along Coupled Microstrip Lines, s = 10 mm

As we can see from the graph, by increasing the spacing of the lines we significantly

reduced the crosstalk effect, in fact also the coupling is lower, and we can also analyze the fact that the voltage on line 1 has a much more similar pattern to the one in Problem nr.1.

A Appendix

A.1 Problem nr.1 MATLAB code

```
clear; clc; close all;

disp('====================================================');
disp('   MICROSTRIP SYNTHESIS AND VOLTAGE EVALUATION');
disp('====================================================');

%% Parameters
er = 9.8;                   % Relative permittivity
h = 1.6e-3;                 % Substrate thickness
Z0 = 50;
f = 10e9;
c = 3e8;

% Input conditions at z = 0
V0 = 10e-3;                 % Input voltage (10 mV)
I0 = 0.4e-3;                % Input current (0.4 mA)
L = 100e-3;                 % Total length of the line (100 mm =
    0.1 m)

%% Microstrip Synthesis
A = (Z0 / 60) * sqrt((er + 1) / 2) + ((er - 1) / (er + 1)) *
    (0.23 + 0.11 / er);

% W/h ratio
W_h = (8 * exp(A)) / (exp(2 * A) - 2);

% Width in meters
W = W_h * h;

% Effective dielectric constant
```

```
eeff = ((er + 1) / 2) + ((er - 1) / 2) * (1 / sqrt(1 + 12 / W_h
    ));

% Display the synthesized microstrip parameters
fprintf('Target Impedance (Z0) : %.1f Ohms\n', Z0);
fprintf('Calculated Parameter A: %.4f\n', A);
fprintf('Calculated W/h ratio  : %.4f\n', W_h);
fprintf('Physical Width (W)    : %.3f mm\n', W * 1000);
fprintf('Effective Permittivity: %.4f\n\n', eeff);

%% Calculate Phase Constant
lambda_0 = c / f;
lambda_g = lambda_0 / sqrt(eeff); % Guide wavelength (m)
beta = 2 * pi / lambda_g;         % Phase constant (rad/m)

fprintf('Guide Wavelength (Lg) : %.2f mm\n', lambda_g * 1000);
fprintf('Phase Constant (beta) : %.2f rad/m\n', beta);
disp('==================================================');

%% Evaluate Voltage along the line
z = linspace(0, L, 1000);

% By using the telegrapher's equation we compute voltage
V_z = V0 * cos(beta * z) - 1j * I0 * Z0 * sin(beta * z);

% Convert everything to mV for a cleaner plot
V_mag_mV = abs(V_z) * 1000;
z_mm = z * 1000;

%% Plot
figure;
plot(z_mm, V_mag_mV, 'b-', 'LineWidth', 2);
grid on;
title('Voltage Magnitude |V_{iso}(z)| along the Microstrip Line
    ');
xlabel('Distance along the line z (mm)');
ylabel('Voltage Magnitude (mV)');
xlim([0, 100]);
ylim([0, max(V_mag_mV) * 1.2]);
```

```
% Highlight the max and min points of the standing wave
hold on;
yline(max(V_mag_mV), 'r--', sprintf('Max Voltage (%.1f mV)',
    max(V_mag_mV)));
yline(min(V_mag_mV), 'g--', sprintf('Min Voltage (%.1f mV)',
    min(V_mag_mV)));
```

A.2 Problem nr.2 MATLAB code

This is only the code for the second part of Problem nr.2, to get the first results shown in the report, change the values of Z_{even}, Z_{odd}, k_{even}, k_{odd}.

```
clear; clc; close all;

%% Initial parameters
er = 9.8;
h = 1.6e-3;
Z0 = 50;
f = 10e9;
c = 3e8;

% Input conditions at z = 0
V1 = 10e-3;
I1 = 0.4e-3;
V2 = 0;
I2 = 0;
L = 100e-3;

% Parameters calculated precedently or gotten from the online
    calculator
eeff = 6.6063;
lambda_0 = c / f;
lambda_g = lambda_0 / sqrt(eeff);
beta = 2 * pi / lambda_g;

Z_even = 50.2845;
Z_odd = 49.0469;
k_even = 6.68437;
k_odd = 6.47581;
```

```
%% Calculations
%  Calculate phase constants
beta_e = (2 * pi * f / c) * sqrt(k_even);
beta_o = (2 * pi * f / c) * sqrt(k_odd);

% Decompose into Even and Odd modes at z = 0
Ve_0 = (V1 + V2) / 2;
Ie_0 = (I1 + I2) / 2;

Vo_0 = (V1 - V2) / 2;
Io_0 = (I1 - I2) / 2;

z = linspace(0, L, 1000);

% Telegrapher's equations for Even and Odd modes
Ve_z = Ve_0 * cos(beta_e * z) - 1j * Ie_0 * Z_even * sin(beta_e
    * z);
Vo_z = Vo_0 * cos(beta_o * z) - 1j * Io_0 * Z_odd * sin(beta_o
   * z);

% 4. Recombine to get the actual physical voltages on Line 1
   and Line 2
V1_z = Ve_z + Vo_z;
V2_z = Ve_z - Vo_z;

% Convert magnitudes to mV for plotting
V1_mag_mV = abs(V1_z) * 1000;
V2_mag_mV = abs(V2_z) * 1000;
z_mm = z * 1000;

%% |Viso| for comparison
Viso_z = V1 * cos(beta * z) - 1j * I1 * Z0 * sin(beta * z);
V_mag_mV = abs(Viso_z) * 1000;
z_mm = z * 1000;

%% Plot
figure;
plot(z_mm, V1_mag_mV, 'b-', 'LineWidth', 2); hold on;
```

```
plot(z_mm, V2_mag_mV, 'r--', 'LineWidth', 2); hold on;
plot(z_mm, V_mag_mV, 'm-.', 'LineWidth', 1);
grid on;
title('Voltage Magnitudes Along Coupled Microstrip Lines');
xlabel('Distance z (mm)');
ylabel('Voltage Magnitude (mV)');
legend('|V_1(z)|', '|V_2(z)|', '|V_{iso}(z)| for comparison');
xlim([0, 100]);
```